Scenario S2-O2-I1 — Our Level 2 (Ideal) / Investor Level 1 (Bootstrapped)

Scenario Summary

Ideal-effort founding team meets a bootstrapped investor. The founders self-fund at full ideal intensity (~~$17,707/mo burn) through Pre-Pilot and Pilot 1, then a bootstrapped-level investor arrives during the Pilot 1 to Pilot 2 transition (approximately M7–M9). The investor contributes just enough to extend runway through Pilot 2 and early Production, but at constrained capacity (~~$7,123/mo effective contribution). This creates a scenario where strong early execution is followed by a resource-constrained scaling phase, lengthening the path to Production and tightening margins during the most operationally demanding stages.

- **Economic cost to Production:** $120,000-155,000 / INR 9,960,000-12,865,000
- **Target timeline to Production:** 26-32 months
- **Founder/company capital before investor:** $71,000-95,000 / INR 5,893,000-7,885,000
- **Investor capital modeled:** $49,000-60,000 / INR 4,067,000-4,980,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Main risk to watch:** bootstrapped investor capital barely covers Pilot 2 + Production staffing ramp; strong Pre-Pilot and Pilot 1 momentum may stall when investor funds arrive at constrained levels

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Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M5	5 months	Architecture freeze (M2), APK backlog lock (M3), doctor training draft (M3), admin callback SOP (M4), internal rehearsal cycles (M4-M5).	Core web flows stable, internal rehearsals pass, doctor and admin runbooks versioned. APK core journeys functional.
Pilot 1	M5-M7	2 months	Warm-network onboarding, first paid consults, callback calibration, doctor playbook iteration, investor pitch deck finalized.	Ethiopia pilot runs live for 2 months with measured throughput; investor term sheet signed.
Pilot 2	M7-M16	9 months	Kenya prep (M12), partner reporting (M13), roster planning, partial automation, staffing calibration, refresher training.	5k-10k user-capable operations with stronger APK, cleaner reporting, staffed callback coverage.
Production Readiness	M16-M28	12 months	Compliance closeout (M20), release freeze (M23), launch playbooks (M25), certification refreshers, production support schedule.	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at ideal pace; APK build running in parallel with web work; doctor/admin training materials drafted; support desk scripts and callback trees defined. Good iteration velocity from L2 resourcing.
Pilot 1	Patient app covers core journeys (registration, dashboard, basic consult); provider web is stable for GP and basic specialist flow; doctor roster, admin callbacks, and manual fulfilment visible in-system. Feature set broader than L1 due to ideal Pre-Pilot investment.
Pilot 2	Core patient features complete; partner dashboards functional; staffing ladder active but at minimum levels due to bootstrapped investor. Reporting and payment loops operational but manual-heavy. Kenya prep begins late in phase. Progress slower than Pre-Pilot/Pilot 1 velocity.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Extended timeline allows thorough but slow hardening due to constrained investor capital.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	96.0%	Next business day	3 core team at ideal utilization + 7 part-time developers at ~480 hrs/mo + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	97.0%	Same-day for critical issues	Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Friendly clinics supplement.
Pilot 2	97.5%	4-hour P1 response	Minimum employed coverage: 3 doctors, 3 admin ops, 2 technical support. Investor capital constrains faster scaling.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 5 doctors, 5 admin ops, 3 technical support for near-24/7 coverage. Below ideal staffing ladder.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	$53,200 / INR 4,415,600
Operating expenditure before Production launch	$84,300 / INR 6,996,900
Total economic spend to Production	**$137,500 / INR 11,412,500**

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	$31,680	INR 2,629,440	3 core team at $20/hr x ~88 hrs/mo x 100% utilization pre-investor = ~$5,280/mo for 5 months Pre-Pilot + Pilot 1. Post-investor at ~60% effective = ~$3,168/mo for 21 months.
Web platform hardening	$14,900	INR 1,236,700	745 hours at $20/hr senior rate. Completed largely during ideal-funded Pre-Pilot and Pilot 1 phases.
Patient APK build	$7,200	INR 597,600	Fixed Android quote: 480 hours at $15/hr. Runs in parallel with web work during Pre-Pilot.
QA, DevOps, security	$6,600	INR 547,800	~330 hours at $20/hr. Adequate scope from ideal Pre-Pilot; constrained post-investor.
Admin and care operations	$86,940	INR 7,216,020	Employed doctors, admin callback staff, and technical support ladder across 23 months of live operations. See ops staffing breakdown below.
Partner onboarding and provider success	$7,500	INR 622,500	Doctor/admin training protocols, SOP writing, partner reporting. Good Pre-Pilot foundation from L2 investment.
Legal, accounting, compliance	$4,200	INR 348,600	Friendly-network pricing at economic value.
Infrastructure, phones, and tools	$24,480	INR 2,031,840	Hosting, devices, telecom/call costs, operational tooling across 28 months. Starts at ideal level, scales more slowly post-investor.
Marketing, travel, rehearsals	$7,000	INR 581,000	Warm-network driven with modest paid acquisition in Pilot 2.
Contingency and buffer	$27,000	INR 2,241,000	~20% buffer on total. Tighter than S2-O1-I1 due to better Pre-Pilot foundation.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M5)	5 months	$0	INR 0	$0
Pilot 1 (M5-M7)	2 months	$2,482	INR 206,006	$4,964
Pilot 2 (M7-M16)	9 months	$3,482	INR 289,006	$31,338
Production Readiness (M16-M28)	12 months	$5,542	INR 459,986	$66,504
Total operational staffing				**$102,806**

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	5 months	$100	INR 8,300	$500
Pilot 1	2 months	$395	INR 32,785	$790
Pilot 2	9 months	$460	INR 38,180	$4,140
Production Readiness	12 months	$750	INR 62,250	$9,000
One-time equipment (phones, laptops)	—	—	—	$1,000
Total infrastructure				**$15,430**

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~100% effective utilization pre-investor	Single owner/founder + 2 founding members. Full parallel execution during Pre-Pilot.
Technical delivery (pre-investor)	7 part-time developers, modeled at ~480 productive hrs/mo	100% of ideal capacity. Strong parallelism between web and APK tracks.
Technical delivery (post-investor)	7 part-time developers, modeled at ~240 productive hrs/mo	Bootstrapped investor constrains team utilization to ~50% of ideal. Significant slowdown.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 2 doctors / Pilot 2: 3 / Production: 5	Below ideal ladder due to constrained investor capital. Partner clinics critical.
Employed admin ops	Pilot 1: 2 admins / Pilot 2: 3 / Production: 5	Callbacks, exception handling, manual fulfilment.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M7)	~$14,830	~INR 1,230,890	100% of full rate card. Ideal execution.
Post-investor (M7-M28)	~$7,400	~INR 614,200	~50% of full rate card. Bootstrapped investor constrains team hours.

Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Basic referral flow live	Grows after referral stabilizes	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	$3.5 / ETB 470	$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by late Production.
Specialist consult	$9 / ETB 1,250	$12 / KES 1,560	Pilot 2 onward.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	$10-15	$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	$9	$14	$20
Gross margin	38%	47%	55%
Monthly ARPU	$1.8	$2.8	$4.2
Monthly churn	13%	11%	8%
LTV/CAC target	1.4x	2.2x	3.0x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M5)	$16,700	INR 1,386,100	2.2 months	Trigger contingency if cash drops below 2 months of current burn.
Pilot 1 (M7)	$15,300	INR 1,269,900	1.4 months	Investor close must happen by M8.
Pilot 2 (M16)	$33,600	INR 2,788,800	3.1 months	Trigger contingency if cash drops below 3 months of burn.
Production Readiness (M28)	$21,400	INR 1,776,200	2.8 months	Trigger contingency if cash drops below 3 months of burn.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	$11,400	$0	$0	$8,230	$3,170	$3,170
M2	Pre-Pilot	$11,400	$0	$0	$8,230	$3,170	$6,340
M3	Pre-Pilot	$11,400	$0	$0	$8,230	$3,170	$9,510
M4	Pre-Pilot	$11,400	$0	$0	$8,230	$3,170	$12,680
M5	Pre-Pilot	$11,400	$0	$0	$9,380	$2,020	$14,700
M6	Pilot 1	$11,400	$0	$500	$11,312	$588	$15,288
M7	Pilot 1	$11,400	$0	$900	$11,312	$988	$16,276
M8	Pilot 2	$0	$54,500	$1,400	$10,561	$45,339	$61,615
M9	Pilot 2	$0	$0	$1,800	$10,561	-$8,761	$52,854
M10	Pilot 2	$0	$0	$2,200	$10,561	-$8,361	$44,493
M11	Pilot 2	$0	$0	$2,600	$10,561	-$7,961	$36,532
M12	Pilot 2	$0	$0	$3,000	$10,561	-$7,561	$28,971

M13	Pilot 2	$0	$0	$3,400	$10,561	-$7,161	$21,810
M14	Pilot 2	$0	$0	$3,800	$10,561	-$6,761	$15,049
M15	Pilot 2	$0	$0	$4,200	$10,561	-$6,361	$8,688
M16	Pilot 2	$0	$0	$4,600	$10,561	-$5,961	$2,727
M17	Prod. Readiness	$0	$0	$5,100	$12,942	-$7,842	-$5,115
M18	Prod. Readiness	$0	$0	$5,800	$12,942	-$7,142	-$12,257
M19	Prod. Readiness	$0	$0	$6,600	$12,942	-$6,342	-$18,599
M20	Prod. Readiness	$0	$0	$7,500	$12,942	-$5,442	-$24,041
M21	Prod. Readiness	$0	$0	$8,500	$12,942	-$4,442	-$28,483
M22	Prod. Readiness	$0	$0	$9,500	$12,942	-$3,442	-$31,925
M23	Prod. Readiness	$0	$0	$10,500	$12,942	-$2,442	-$34,367
M24	Prod. Readiness	$0	$0	$11,500	$12,942	-$1,442	-$35,809
M25	Prod. Readiness	$0	$0	$13,000	$12,942	$58	-$35,751
M26	Prod. Readiness	$0	$0	$14,500	$12,942	$1,558	-$34,193
M27	Prod. Readiness	$0	$0	$16,000	$12,942	$3,058	-$31,135
M28	Prod. Readiness	$0	$0	$18,000	$12,942	$5,058	-$26,077

Note: Negative ending cash from M17 onward indicates that the bootstrapped investor capital is insufficient to fund Production Readiness without additional capital injection. This scenario requires either (a) a bridge round at ~M16, (b) revenue acceleration beyond modeled targets, or (c) cost reductions in Production Readiness staffing. See Key Risks.

Totals:

- Founder capital in: $91,200 / INR 7,569,600 (M1-M7 at ~$11,400 x 7 months + phase adjustments, range $71,000-$95,000)
- Investor capital in: $54,500 / INR 4,523,500 (range $49,000-$60,000)
- Cumulative revenue: ~$154,900
- Cumulative outflow: ~$310,900

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	$145,700 / INR 12,093,100
Total economic cost	$137,500 / INR 11,412,500
Sweat-equity gap (economic cost minus cash invested)	~-$8,200 (cash exceeds modeled economic cost due to contingency absorption)
Cumulative revenue at M28	~$154,900 / INR 12,856,700
Monthly break-even (revenue >= outflow)	~M25 (month 25)
Cumulative break-even (total revenue >= total capital)	~M32-M38 (4-10 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M34-M40 (6-12 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	$91,200 / INR 7,569,600 (range $71,000-$95,000)
Investor round size	$54,500 / INR 4,523,500 (range $49,000-$60,000)
Recommended structure	SAFE or milestone-based bridge note
Modeled pre-money valuation	$272,500
Modeled post-money valuation	$327,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	Drawn from the dedicated investor pool with minor proportional dilution.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
Cash runs out during Production Readiness (~M16-M17)	High	Critical	Bootstrapped investor capital is insufficient for full Production Readiness. Must plan a bridge round by M14 or achieve revenue targets 20% above model.
Velocity drop when investor capital constrains team	High	High	Team accustomed to ideal pace during Pre-Pilot/Pilot 1 will experience 50% reduction in productive hours. Communicate expectations early; re-sequence non-critical work.
Investor does not close by M8	Medium	Critical	Runway extends to M9-M10 on founder capital alone. Pre-negotiate term sheet during Pilot 1.
Manual ops overload at constrained staffing	High	Medium	Pilot 2 and Production staffing is below ideal ladder. Track callback backlog as release gate. Add capacity before feature expansion.
Team morale drop from velocity change	Medium	High	Strong Pre-Pilot velocity creates expectations. When post-investor pace slows, team may disengage. Set phase-specific velocity targets.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Count friendly counsel at economic value.
Payment-rail mismatch	Medium	Medium	Use digital invoicing first; localize M-Pesa / Telebirr before scaling.
Kenya market entry delayed	Medium	Medium	Focus on Ethiopia stability first. Kenya prep is a Pilot 2 activity, not a Pilot 1 blocker.

Recommendation

S2-O2-I1 is a **structurally imbalanced scenario**. The founders invest at ideal levels during Pre-Pilot and Pilot 1, building strong momentum and a quality foundation, but the bootstrapped investor provides insufficient capital to sustain that momentum through Pilot 2 and Production Readiness. The cash flow model shows negative balances from M17 onward, indicating a funding gap that must be addressed.

This path is viable only if:

1. Revenue ramps faster than modeled (reaching $10,000+/mo by M22 instead of M24)
2. A bridge round or follow-on investment is secured during Pilot 2 (M14-M16)
3. The founding team accepts a significant velocity reduction post-investor and can manage team morale through the transition
4. Production Readiness staffing is kept at minimum viable levels (below ideal ladder)

The core tension: ideal founder investment creates quality and momentum that bootstrapped investor capital cannot sustain. This mismatch makes the scenario fragile at the Pilot 2 to Production transition. Consider S2-O2-I2 for a balanced path or S2-O2-I3 for a fully funded scaling phase.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at $15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors ($720/mo), INR 18,000 for admin ops ($217/mo), INR 25,000 for technical support ($301/mo).
- Our L2 (ideal) spend is modeled at 100% of team utilization, yielding ~$14,830/mo team cost + ~$2,877/mo ops/infra/overhead = ~$17,707/mo total burn pre-investor.
- Investor L1 (bootstrapped) contribution sized at ~$49,000-60,000, representing bare-minimum capital addition. Effective contribution ~$7,123/mo when active.
- Pre-investor founder contribution: ~$11,400/mo ($8,230 team/infra + $3,170 buffer for ops/overhead ramp).
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.

End of S2-O2-I1 scenario. All assumptions trace to params.md v2.